Natural selection

Click on **Interactives**.

Click on **Interactive Modules**.

Click on **Natural Selection**.

Work through the interactive module to find the answers to complete this worksheet.

Key terms
adaptation, evolution, characteristics, environment, extinct, genes, genetic variation, gene pool, predators, reproduction, selection, survival

Missing words are not necessarily listed in the Key Terms.

Causes of natural selection

Natural selection is the mechanism of _____ change. Natural selection relies on the following conditions:

1 _____ exists within a population of organisms.

2 All organisms struggle to _____ and _____. For example, it might be difficult to find _____ , _____ or _____ or to avoid _____ .

3 The organisms that are best suited to their environment will _____ and _____ .

4 The organisms that survive and reproduce pass their _____ and genetic information on to their offspring.

5 Over time and many generations the population will become _____ _____ to its environment.

Natural selection can lead to a _____ in the characteristics of populations living in a changing environment, or can lead to _____ of characteristics when the environment stays constant.

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Adelie penguins

Now let's explore how natural selection can affect a population of Adelie penguins.

1 On which continent would you find Adelie penguins? _____

2 The characteristics of Adelie penguins:

a They feed at _____ .

b They _____ and _____ .

c They keep the same _____ each year.

d They keep the same _____ _____ each year.

e They lay _____ eggs each year.

f The partners take turns to _____ the eggs.

Variation

1 Characteristics that may vary within a population of penguins include _____ , _____ , _____ _____ , and the amount of _____ .

2 Think of two other characteristics that may vary in the individuals of penguin population.

a _____

b _____

3 Offspring are similar to their parents because they _____ their characteristics in the form of _____ (the genetic information that controls the characteristics). Because life is tough, some of the chicks will not survive to pass on their genes.

Simulator

1 What is blubber?

_____

_____

_____

2 Why is blubber an advantage to the survival of Antarctic penguins?

_____

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_____

_____

_____

_____

3 Run the simulator to find out what sort of chick is born with each of the following penguin matings. Click the BREED button and **watch carefully** to see the offspring from each mating.

a fat × fat → _____

b fat × medium → fat or _____

c fat × thin → _____

d medium × medium → _____ or _____ or _____

e medium × thin → medium or _____

f thin × thin → _____

If you want to watch it again, click Restart Experiment

This pattern of genetic inheritance is called **co-dominance** because when fat and thin penguins mate these characteristics do not dominate over each other. Instead they have an intermediate offspring.

4 Use the simulator to make 5 generations of penguins under 'Normal' environmental conditions. After each new generation of chicks is born, some of them die because they are not best adapted. This is what the **cull** means.

Each simulation starts with 10 penguins of each characteristic type: fat, medium, and thin. Complete the table below by recording the number of each type of penguin born and how many are left after the cull in Normal environmental conditions.

Environment is Normal	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born			
Cull condition: Normal			
Generation 2 born			
Cull condition: Normal			
Generation 3 born			

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Cull condition: Normal			
Generation 4 born			
Cull condition: Normal			
Generation 5 born			
Cull condition: Normal			

When the environmental conditions stayed the same at the 'Normal' Antarctic conditions, what was the most common thickness of blubber on the penguins after 5 generations?

_____

5 Now that you can use the simulator, experiment with different environmental conditions and discover the effect on the penguin population. Here are some things you could try.

a Breed 5 generations of penguins with the temperature getting colder each year. When you cull, select 'colder' or 'much colder'.

Environment is _____	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born			
Cull condition: _____			
Generation 2 born			
Cull condition: _____			
Generation 3 born			
Cull condition: _____			
Generation 4 born			
Cull condition: _____			
Generation 5 born			
Cull condition: _____			

After 5 generations what is the most common thickness of blubber?

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_____

b Breed 5 generations of penguins with the temperature getting warmer each year. When you cull, select ‘warmer’ or ‘much warmer’.

Environment is _____	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born			
Cull condition: _____			
Generation 2 born			
Cull condition: _____			
Generation 3 born			
Cull condition: _____			
Generation 4 born			
Cull condition: _____			
Generation 5 born			
Cull condition: _____			

After 5 generations what is the most common thickness of blubber?

_____

c Breed 5 or more generations of penguins with the temperature changing in whatever way you like.

Environment is _____	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born			
Cull condition: _____			
Generation 2 born			
Cull condition: _____			

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Generation 3 born			
Cull condition: _____			
Generation 4 born			
Cull condition: _____			
Generation 5 born			
Cull condition: _____			
Generation 6 born			
Cull condition: _____			
Generation 7 born			
Cull condition: _____			
Generation 8 born			
Cull condition: _____			
Generation 9 born			
Cull condition: _____			
Generation 10 born			
Cull condition: _____			

Describe the effects of the environmental changes on the characteristics of your penguin population:

_____

_____

_____

_____

_____

_____

_____

The gene pool

1 The genetic information, or all the **genes**, within a whole population of organisms is called the _____ _____ .

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2 An organism that has a combination of genes that allows it to survive when the environment changes is called well _____ to the environment and its characteristics are called _____ .

3 If conditions change and no individuals survive long enough to reproduce, then that species becomes _____ .

Questions

Complete this matching task from the information in the interactive module. Write the correct number from the left-hand column next to the matching meaning or example in the right-hand column.

Term	**Meaning or example**
1 Natural Selection	___ An organism can find enough food, shelter and a mate
2 Charles Darwin	___ Two organisms produce offspring
3 Evolution	___ A physical or behavioral feature of an organism
4 Genetic variation	___ A unit of genetic information that determines a characteristic of an organism, e.g. feather colour
5 Survival	___ Adelie penguins with thin blubber do not survive long enough to reproduce in extremely cold conditions
6 Reproduction	___ Within a population of finches (a bird) some have thick beaks and some have thinner beaks. Beak thickness is determined by a gene
7 Characteristic	___ Loss of all the genetic information of a species because no individuals survive long enough to reproduce
8 Predator	___ All of the genes or genetic information within a population
9 Environment	___ Cold, hot, humid or dry climate, flat or mountainous terrain, forest or savanna or desert are all part of this
10 Gene	
11 Gene pool	
12 Extinction	

Now answer the questions on the last screen of the interactive module.

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Natural selection: Answers

Click on **Interactives**.

Click on **Interactive Modules**.

Click on **Natural Selection**.

Work through the interactive module to find the answers to complete this worksheet.

Key terms
adaptation, evolution, characteristics, environment, extinct, genes, genetic variation, gene pool, predators, reproduction, selection, survival

Missing words are not necessarily listed in the Key Terms.

Causes of natural selection

Natural selection is the mechanism of *evolutionary* change. Natural selection relies on the following conditions:

1 *Genetic variation* exists within a population of organisms.

2 All organisms struggle to *survive* and *reproduce*. For example, it might be difficult to find *food*, *shelter* or *a mate* or to avoid *predators*.

3 The organisms that are best suited to their environment will *survive* and *reproduce*.

4 The organisms that survive and reproduce pass their *characteristics* and genetic information on to their offspring.

5 Over time and many generations the population will become *better adapted* or *better suited* to its environment.

Natural selection can lead to a *change* in the characteristics of populations living in a changing environment, or can lead to *stability* of characteristics when the environment stays constant.

Adelie penguins

Now let's explore how natural selection can affect a population of Adelie penguins.

1 On which continent would you find Adelie penguins? *Antarctica*

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2 The characteristics of Adelie penguins:

a They feed at *sea*.

b They *swim* and *dive*.

c They keep the same *mate* each year.

d They keep the same *nest site* each year.

e They lay *2* eggs each year.

f The partners take turns to *incubate* the eggs.

Variation

1 Characteristics that may vary within a population of penguins include *size*, *colour*, *beak shape or size*, and the amount of *blubber*.

2 Think of two other characteristics that may vary in the individuals of penguin population.

Examples include:

a *swimming ability; diving ability*

b *thickness of feathers (insulation); amount of oil on outer feathers (waterproofing)*

3 Offspring are similar to their parents because they *inherit* their characteristics in the form of *genes* (the genetic information that controls the characteristics). Because life is tough, some of the chicks will not survive to pass on their genes.

Simulator

1 What is blubber?

The layer of fat beneath the skin that insulates the penguins from the cold.

2 Why is blubber an advantage to the survival of Antarctic penguins?

Because fat is an insulator. It helps keep the internal temperature of the penguins high despite the low external temperature.

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_____

_____

3 Run the simulator to find out what sort of chick is born with each of the following penguin matings. Click the BREED button and **watch carefully** to see the offspring from each mating.

a fat × fat → *fat*

b fat × medium → fat or *medium*

c fat × thin → *medium*

d medium × medium → *fat* or *medium* or *thin*

e medium × thin → medium or *thin*

f thin × thin → *thin*

If you want to watch it again, click Restart Experiment

This pattern of genetic inheritance is called **co-dominance** because when fat and thin penguins mate these characteristics do not dominate over each other. Instead they have an intermediate offspring.

4 Use the simulator to make 5 generations of penguins under 'Normal' environmental conditions. After each new generation of chicks is born, some of them die because they are not best adapted. This is what the **cull** means.

Each simulation starts with 10 penguins of each characteristic type: fat, medium, and thin. Complete the table below by recording the number of each type of penguin born and how many are left after the cull in Normal environmental conditions.

Environment is Normal	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born	*13*	*18*	*14*
Cull condition: Normal	*8*	*13*	*9*
Generation 2 born	*11*	*22*	*12*
Cull condition: Normal	*6*	*17*	*7*
Generation 3 born	*9*	*25*	*11*
Cull condition: Normal	*4*	*20*	*6*

Numbers will vary with every simulation but the trend should be consistent.

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Generation 4 born	*7*	*26*	*12*
Cull condition: Normal	*2*	*21*	*7*
Generation 5 born	*5*	*26*	*14*
Cull condition: Normal	*0*	*21*	*9*

When the environmental conditions stayed the same at the 'Normal' Antarctic conditions, what was the most common thickness of blubber on the penguins after 5 generations?

medium

5 Now that you can use the simulator, experiment with different environmental conditions and discover the effect on the penguin population. Here are some things you could try.

a Breed 5 generations of penguins with the temperature getting colder each year. When you cull, select 'colder' or 'much colder'.

One example:

Environment is *much colder*	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born	*14*	*16*	*15*
Cull condition: *much colder*	*14*	*16*	*0*
Generation 2 born	*23*	*22*	*0*
Cull condition: *much colder*	*23*	*7*	*0*
Generation 3 born	*35*	*9*	*1*
Cull condition: *much colder*	*30*	*0*	*0*
Generation 4 born	*43*	*2*	*0*
Cull condition: *much colder*	*30*	*0*	*0*
Generation 5 born	*44*	*1*	*0*
Cull condition: *much colder*	*30*	*0*	*0*

After 5 generations what is the most common thickness of blubber?

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fat

b Breed 5 generations of penguins with the temperature getting warmer each year. When you cull, select 'warmer' or 'much warmer'.

One example:

Environment is *much warmer*	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born	*15*	*16*	*14*
Cull condition: *much warmer*	*0*	*16*	*14*
Generation 2 born	*2*	*21*	*22*
Cull condition: *much warmer*	*0*	*8*	*22*
Generation 3 born	*0*	*11*	*34*
Cull condition: *much warmer*	*0*	*0*	*30*
Generation 4 born	*0*	*1*	*44*
Cull condition: *much warmer*	*0*	*0*	*30*
Generation 5 born	*0*	*0*	*45*
Cull condition: *much warmer*	*0*	*0*	*30*

After 5 generations what is the most common thickness of blubber?

thin

c Breed 5 or more generations of penguins with the temperature changing in whatever way you like.

This activity is intended for exploration. One example:

Environment is *normal → colder → warmer*	**Number of penguins**		
	Fat	**Medium**	**Thin**
Generation 0	10	10	10
Generation 1 born	*14*	*17*	*14*
Cull condition: *normal*	*9*	*12*	*9*

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Generation 2 born	*12*	*21*	*12*	
Cull condition: *colder*	*12*	*18*	*0*	
Generation 3 born	*19*	*23*	*3*	
Cull condition: *colder*	*19*	*11*	*0*	
Generation 4 born	*27*	*17*	*1*	
Cull condition: *much colder*	*27*	*3*	*0*	
Generation 5 born	*40*	*5*	*0*	
Cull condition: *normal*	*30*	*0*	*0*	
Generation 6 born	*45*	*0*	*0*	
Cull condition: *normal*	*30*	*0*	*0*	
Generation 7 born	*44*	*1*	*0*	
Cull condition: *warmer*	*30*	*0*	*0*	
Generation 8 born	*43*	*2*	*0*	
Cull condition: *much warmer*	*0*	*0*	*0*	*EXTINCTION!*
Generation 9 born				
Cull condition: _____				
Generation 10 born				
Cull condition: _____				

Describe the effects of the environmental changes on the characteristics of your penguin population:

As climate cools, more fat penguins survive. As climate warms, more thin penguins survive.

In the example given, the population was unable to adapt to the sudden climate change (warming) because there were not enough penguins with the genes for thin blubber. Extinction occurred.

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The gene pool

1 The genetic information, or all the **genes**, within a whole population of organisms is called the *gene pool*.

2 An organism that has a combination of genes that allows it to survive when the environment changes is called well *adapted* to the environment and its characteristics are called *adaptations*.

3 If conditions change and no individuals survive long enough to reproduce, then that species becomes *extinct*.

Questions

Complete this matching task from the information in the interactive module. Write the correct number from the left-hand column next to the matching meaning or example in the right-hand column.

Term	**Meaning or example**
1 Natural Selection	*5* An organism can find enough food, shelter and a mate
2 Charles Darwin	*6* Two organisms produce offspring
3 Evolution	*7* A physical or behavioral feature of an organism
4 Genetic variation	*10* A unit of genetic information that determines a characteristic of an organism, e.g. feather colour
5 Survival	*1* Adelie penguins with thin blubber do not survive long enough to reproduce in extremely cold conditions
6 Reproduction	*4* Within a population of finches (a bird) some have thick beaks and some have thinner beaks. Beak thickness is determined by a gene
7 Characteristic	*12* Loss of all the genetic information of a species because no individuals survive long enough to reproduce
8 Predator	*11* All of the genes or genetic information within a population
9 Environment	*9* Cold, hot, humid or dry climate, flat or mountainous terrain, forest or savanna or desert are all part of this
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Now answer the questions on the last screen of the interactive module.

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